



Birla Institute of Technology & Science, Pilani

Pilani Campus

AUGS/AGSR Division

GPIC 2026 | Gujarat Police Innovation Challenge

Module 01 — PRAHARI

Lectures Hackathon demo

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Outcomes

- Recite the official test case: onboard the grid, track a plate, raise a watchlist alert
- Name the hybrid (Models 1 + 2 + thin 3) and say why Model 4 is Phase-2
- State the MEASURED live result: 30 catalogue cameras, RTSP-TCP frame, cam04 confirm
- Separate MEASURED laptop facts from DESIGN TARGET 80k / 5-6 Cr numbers



Official challenge (source)

- Home Department, Government of Gujarat. CCTV Integration Hackathon 2026
- Page: <https://sentinel.gujarat.gov.in/problems>
- Category 1 student team. Phase 1 lock 07 September 2026, 12:00 IST
- Transcript of every word on that page: 06_References/SENTINEL_Problems_Page.md



What the State already has

- 26 departments run independent CCTV estates (analog and IP)
- Sites from border districts to Valsad, Dahod, Somnath, Jamnagar, Dwarka
- Cloud NVR in some estates, local storage in others; 7-day vs 15-day retention
- Watchlists already exist: VAHAN, SARTHI, eGujCop, AFIS, NAFIS
- The cameras do not match against those lists. There is no statewide census



Four official challenges

- Heterogeneous infrastructure: vendors, VMS, AMC, formats, protocols
- Geographical dispersion of about 1,000 km
- Unified analytics and event handling across onboarded cameras
- New cameras and departments without a redesign

Core goal on the official page: use existing infrastructure to the maximum practical extent.



Reference models (official)

Model	Official name	What it does
1	Registry and GIS	Census, health, gaps. No central recording
2	Unified viewing	Direct RTSP / ONVIF / API + ANPR
3	VMS federation	Middleware, adapters, event bus
4	Central VMS	Ingest, store, playback, full AI
H	Hybrid	Combine two or more

- Model 1 is mandatory foundation for 2, 3, and 4



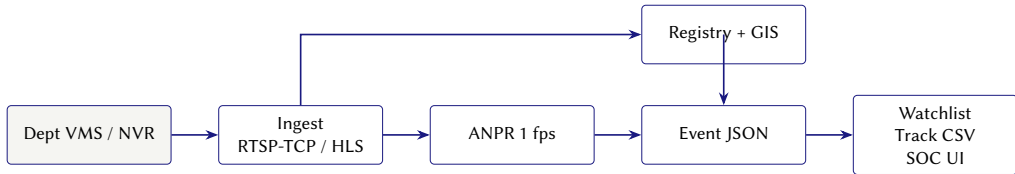
Chosen model

Layer	Official model	PoC
Census, GIS, health, gaps	1	now
Unified viewing + ANPR	2	now
Event bus, watchlist, track	thin 3	now
Central VMS recording	4	Phase-2, selected cameras

- A new vendor is one adapter on RTSP / HLS / WHEP / ONVIF
- Detection JSON does not change if the bus later becomes Kafka
- Model 4 on day one would fake a VMS rip-out this team cannot host



Logical architecture



- Analytics samples 1 frame per second, not a 25 fps GOP
- PoC bus is an in-process WebSocket; the JSON is the contract



Official test case

- Onboard about 50 heterogeneous cameras from the Resources page
- Integrate live feeds. Enable monitoring and video analytics
- Identify a designated vehicle registration number across the grid
- Reconstruct route, timestamps, locations. Plot on GIS
- Match live detections to a representative watchlist. Raise automated alerts
- Output: working platform, not a mock. CSV / report with timestamps



Live Sentinel contract (03 Sep 2026)

- Portal: <https://cctv.corp8.cloud/> after access-password login
- Catalogue: GET /cameras.json returns 30 rows {id, name}
- /api/ingest is HTTP 404 on this host
- HLS: <https://cctv.corp8.cloud/<id>/index.m3u8> (cookie + browser UA)
- RTSP: <rtsp://103.250.160.189:8554/stream/<id>> (TCP). TLS host does not pass 8554
- One second of video takes one second to arrive. There is no file download



Integrator laws in the capture client

- Set `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` before `import cv2`
- Event time is PTS (`CAP_PROP_POS_MSEC`), never arrival time, never declared FPS
- Reconnect backoff starts at 2 s and caps at 30 s
- Decoder warnings at join are logged, not fatal
- PTS rewind or gap > 5 s is a scene cut (feeds loop)
- Consume only. Do not publish. Do not `wget /stream/<id>`
- Tests: `tests/test_integrator_laws.py`



MEASURED: government feed (03 Sep 2026)

- Synced 30 catalogue cameras into the registry (cam01 ...cam30)
- RTSP-TCP probe live: cam04 Paldi Circle, cam01 Chimanbhai Bridge
- First live frame: 03_Data/recordings/first_live_frame.png
- Size 1920×1080, PTS 1080 ms. UDP SETUP 461 then TCP succeeds
- Manifest has no live flag and no lat/lon. Pins stay off the map until GIS arrives
- Open tiles from the Cameras table, not from a 0,0 marker



MEASURED: designated vehicle

Time	Camera	Location
06:12	CAM-VAL-001	Valsad NH-48 Toll
07:48	CAM-SUR-001	Surat Adajan Circle
09:05	CAM-AHD-004	Narol-Naroda Hwy
09:22	CAM-AHD-001	SG Highway Junction
09:51	CAM-GNR-003	Koba Circle
10:04	CAM-GNR-001	CH-0 Sector 21 GNR
11:40 IST	cam04	Paldi Circle (live confirm)

- Plate GJ01AB1234, watchlist WL-001 / STOLEN
- Live confirm appends. It does not drop the six seed points
- Track length MEASURED: 8 cameras. CSV: gov_feed_plates.csv



ANPR and operator confirm

- Sampled frame → plate box → Tesseract → Indian normaliser
- Regex after stripping space and hyphen: `^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$`
- Interface: `recognize(frame) -> {plate, plate_raw, confidence, crop}`
- Operator confirm writes confidence 1.0 so a readable plate cannot fail the demo
- MEASURED 03 Sep: Tesseract binary absent. Confirm path used on cam04
- Production swap (same function): YOLO plate detector + PaddleOCR



Watchlist and alerts

- In-memory plate set, $O(1)$ match, reload every 60 s
- Representative sources: VAHAN, SARTHI, eGujCop. No live ministry API in the PoC
- CRITICAL stolen/wanted (red); HIGH (amber); LOW (log)
- Same plate + same camera within 120 s increments a counter
- WebSocket /ws/alerts plus 5 s poll fallback
- MEASURED: confirm on cam04 raised a CRITICAL alert for GJ01AB1234



Evaluation map A (official areas 1–4)

Official area	PRAHARI now
1 Test case on gov feed	30 cameras onboarded. RTSP-TCP frame. Confirm + CSV on cam04
2 Solution presentation	This deck + PDF. Hybrid justified
3 Architecture	HLD. Open adapters. Model 4 not faked
4 Working platform	FastAPI on :8080. Own-feed file + live catalogue



Evaluation map B (official areas 5–7 and bonus)

Official area	PRAHARI now
5 Analytics output	Confirm row + timestamps. OCR pending Tesseract on PATH
6 Scale / PoC ready	1 fps regional design target in HLD §5 / §10
7 Completeness	GitHub live. YouTube and Drive still human
Bonus (on camera)	Hybrid, track, RBAC, gaps, WS alerts, 1 fps cost sentence



Eighty thousand cameras (DESIGN TARGET)

- About 45,000 public-domain cameras run analytics at 1 fps, 720p crop ≈ 80 KB
- Naive centre: $45,000 \times 80 \text{ KB/s} \approx 3.6 \text{ GB/s}$
- Five regions (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj): $\approx 720 \text{ MB/s}$ each
- SOC wall pulls 16–64 HLS tiles. Everything else is metadata + crops
- These numbers are design targets, not measured laptop throughput



Cost of the intelligence plane (DESIGN TARGET)

Item (statewide, per year)	INR
5 regional GPU nodes (3-yr amortised)	1.8–2.4 Cr
Object storage for crops + selected video	0.8–1.2 Cr
SWAN / leased bandwidth top-up	0.6–1.0 Cr
Integration + SOC ops (12 engineers)	1.5 Cr
Total intelligence plane	~5–6 Cr

- This is not the bill for replacing 26 VMS estates



Security in the PoC

- Roles: superadmin, soc_operator, dept_viewer, auditor
- home.viewer sees Home cameras only; private-permitted mall is hidden
- HLS/file tiles use a 60 s HMAC token. Raw RTSP never reaches the browser
- Sentinel cookie stays on the server. The browser never sees it
- Private-permitted onboard requires consent=true
- Audit: onboard, watchlist edit, ack, report download, operator confirm



Stack (open source, as required)

Piece	PoC	Later swap
API	FastAPI + Uvicorn	same contract
UI	Leaflet + vanilla JS	React if needed
Store	SQLite	PostgreSQL + PostGIS
Capture	OpenCV + FFmpeg, RTSP TCP	DeepStream
ANPR	morphology + Tesseract	YOLO + PaddleOCR
Alerts	in-process WebSocket	Redis / Kafka

- Tests MEASURED 03 Sep: 33 passed, 1 skipped (Tesseract binary)



What is done, what is still human

- Done: platform, HLD, this deck, live catalogue sync, RTSP frame, confirm CSV
- Done: GitHub <https://github.com/amitduabits/PRAHARI>
- Human: own-feed YouTube ≤ 3 min (script `own_feed_demo_script.md`)
- Human: gov-feed YouTube ≤ 3 min (sync, tile, confirm, CSV)
- Human: Drive link, Anyone + Viewer, for `gov_feed_plates.csv`
- Human: change `JUDGE_PASSWORD`; optional hosted URL
- Stop product edits after the portal receipt on 07 Sep



Demo run (3 minutes)

- Open <http://127.0.0.1:8080>. Login judge
- Operations map: seeded pins, Dahod offline on purpose
- Cameras table: Open tile on cam04 or cam01
- Vehicle Track: reconstruct GJ01AB1234, download CSV
- Alerts: CRITICAL queue, Ack writes audit
- Onboard: Operator confirm if OCR misses
- Gaps: short retention on Food and Civil Supplies